SANDAS WIJETHUNGA

Data Science Undergraduate

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PROFESSIONAL SUMMARY

Data Science undergraduate with a solid foundation in statistics, time series analysis, and machine learning, plus academic exposure to data engineering, financial analytics, and deep learning. Hands-on experience with R, Python, SQL, and Power BI through forecasting, network analytics, and cloud data projects, along with exposure to deep learning frameworks such as TensorFlow, Keras and PyTorch. Aspiring AI researcher, driven to build data-driven solutions with real-world impact.

EDUCATION

BSc (Hons) in Data Science 2024 – 2027

Coventry University | National Institute of Business Management (NIBM)

Secondary Education — GCE Advanced Level (Physical Science Stream) 2008 – 2022

Prince of Wales' College, Moratuwa

TECHNICAL SKILLS

- Machine Learning and AI**:** Supervised Learning, Unsupervised Learning, Reinforcement Learning, Deep Learning, Time Series Analysis, PyTorch, TensorFlow
- Data Engineering and Big Data**:** Data Pipelines, Data Processing, ETL Concepts, Big Data Analytics
- Programming and Tools**:** Python (Pandas, NumPy, Scikit-learn), R, SQL & NoSQL, Cloud Computing
- Data Visualization**:** Power BI, Matplotlib, Seaborn
- Mathematics and Analytics**:** Statistics, Probability, Calculus, Linear Algebra

RESEARCH

A Comparative Evaluation of Statistical and Multivariate AR-Net Models for Sector-Level Electricity Demand Forecasting in Sri Lanka Using Macroeconomic and Climatic Exogenous Variables

- Built and benchmarked five forecasting models (ARIMA, ARIMAX, SARIMA, SARIMAX, Multivariate AR-Net) on 25 years of monthly Sri Lankan electricity consumption data, incorporating macroeconomic and climatic exogenous variables (GDP, population, IIP, temperature, humidity).
- Engineered a dual-language pipeline (R and Python/TensorFlow-Keras) with Box-Cox/log transformations and RobustScaler normalization to handle structural breaks from COVID-19 and the 2022 economic crisis.
- Applied Granger Causality testing and a Random Forest + Mutual Information ranking approach for feature selection, then evaluated all models on an identical 24-month held-out test set.
- Achieved the best results with the Multivariate AR-Net (MAPE of 4.34% domestic, 6.57% industrial), producing 3-month-ahead forecasts for national energy planning.

ACADEMIC PROJECTS

LeafScan AI — Tea Leaf Disease Detection | Deep Learning

- Developed an end-to-end image classifier: trained an EfficientNetB0 transfer learning model using TensorFlow/Keras to classify four tea leaf disease categories, achieving 93.6% test accuracy.
- Deployed a web app and REST API: built a Flask interface for real-time inference, containerized with Docker and orchestrated via Docker Compose for seamless deployment.
- Integrated MLOps and experiment tracking with MLflow, systematically logging parameters, metrics, and model artifacts to ensure full project reproducibility.
- Troubleshot deployment environments, resolving Python and Keras dependency version conflicts between local and containerized environments.

- Evaluated model and data using precision, recall, F1-score, and confusion matrices, documenting critical dataset limitations.

Tech Stack: Python, TensorFlow, Keras, Flask, Docker, MLflow, scikit-learn, REST APIs

CityPulse — Weather Analytics Pipeline | Data Engineering

- Built an end-to-end streaming ELT pipeline: ingested live weather data for 5 cities hourly via REST API, streamed through Kafka into PostgreSQL, and transformed daily with PySpark into a star-schema warehouse.
- Engineered idempotent ingestion using Kafka-to-database loading with offset-commit ordering and database uniqueness constraints, eliminating duplicate ingestion on consumer retries.
- Orchestrated automated workflows with 2 Apache Airflow DAGs configured for automatic retries, running unattended and self-recovering from downtime.
- Containerized the full multi-service application with Docker Compose for single-command deployment.

Tech Stack: Docker, Apache Airflow, Apache Kafka, PySpark, PostgreSQL, REST APIs

Information Retrieval System — Web Crawling, Indexing & Search | Search & Ranking Algorithms

- Built a web crawler and inverted index from scratch, collecting and indexing web pages with tokenization, normalization, and stemming/lemmatization.
- Merged multiple independently built indexes into a single unified collection-wide inverted index, doc-length map, and document ID mapping.
- Implemented the Vector Space Model (VSM) from scratch, using TF-IDF weighting and cosine similarity to rank documents against a query.
- Implemented the Query Likelihood Model (QLM) from scratch, applying smoothed language-model scoring over collection statistics.
- Deployed both retrieval models through a Streamlit search application, allowing side-by-side comparison of VSM and QLM results.

Tech Stack: Python, NLTK, Streamlit, JSON-based inverted index

SUB PROJECTS

- **Human Activity Monitoring System:** Cloud-based IoT monitoring system for heart rate and body temperature using MongoDB Atlas, with custom aggregation pipelines and a live Atlas Charts dashboard.
- **Network Visualization using Neo4j and Bokeh:** Graph-based modeling of the Netflix titles dataset in Neo4j with interactive visualizations built in Bokeh.
- **Survey Dashboard:** Interactive dashboard analyzing survey response data, built with Python and accompanied by a full analysis report.
- **E-Commerce Platform:** Cloud-based e-commerce platform on AWS supporting personalized online ordering.
- **Demographics and Music Engagement Case Study:** Analyzed how demographic factors shape music engagement patterns.

WORK EXPERIENCE

Operations Executive | Commercial Bank March 2023 – March 2024

- Resolved customer issues related to digital banking products, providing technical troubleshooting support via phone and email for online and mobile banking platforms.
- Diagnosed and explained technical problems in clear, non-technical terms to customers with varying levels of digital literacy, ensuring accurate issue resolution and reducing repeat complaints.
- Handled a high volume of customer interactions across multiple channels while maintaining accuracy and compliance with banking operational standards.
- Documented recurring technical issues and customer feedback, contributing to process improvements for the digital banking support workflow.

CERTIFICATIONS

- Databases and SQL for Data Science with Python — IBM | Completed
- AWS Cloud Technical Essentials — Amazon Web Services | In Progress
- Microsoft Power BI: Data Analysis Professional Specialization | In Progress